S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026 · AI/NLP RISK ENGINE CASE STUDY

RiskPulse

Turning news, global events and social posts into structured risk signals, and an index that rebalances on them.



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3
live data sources

569
structured risk signals

74
index rebalances

PROBLEM & APPROACH

Market-moving risk arrives as text

Thousands of headlines, wire stories and posts reach the market every day. They are unstructured, noisy and fast: no analyst can read them all, and a static index cannot react to them.

The brief: an **AI/NLP Risk Engine** that turns this text into machine-readable risk signals, plus a downstream module that acts on them.

Our answer is **RiskPulse**: three steps, each a separate module that can be tested and swapped.

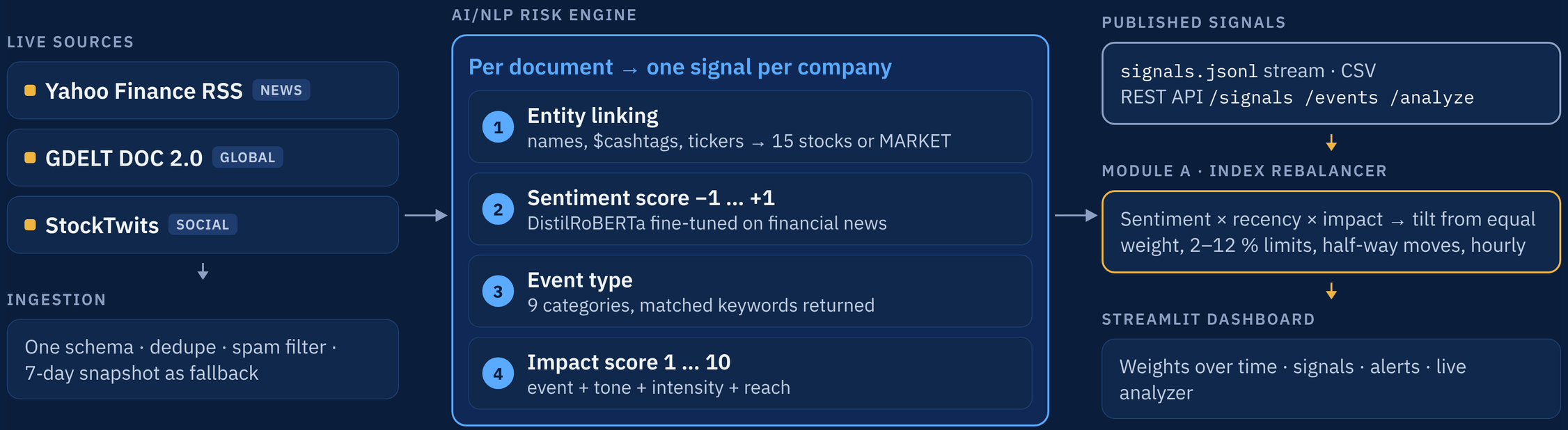
1 Ingest
Yahoo Finance news, GDELT global news and StockTwits posts: live, free, no API keys

2 Structure
Per company: sentiment score −1…+1, event type and impact score 1–10

3 Act
Module A tilts a 15-stock S&P 100 index toward positive and away from negative sentiment, every hour

SYSTEM DESIGN

Architecture and data flow



IMPLEMENTATION HIGHLIGHTS

Key modules, tech choices and why

Module	What it does	Why this choice
Ingestion	3 connectors in parallel · one schema · dedupe · spam & copy-paste filter	Free APIs fail and rate-limit: a 7-day snapshot keeps every cycle running
Sentiment model	DistilRoBERTa fine-tuned on financial news; score = P(pos) – P(neg)	Won our benchmark: macro-F1 0.71 vs FinBERT 0.66, 1.9× faster on CPU
Events & impact	9-type weighted keyword taxonomy; impact = event + tone + intensity + reach	No labeled event data needed; every label shows the keywords behind it
Module A	Recency × impact × source weights, shrinkage, 2–12 % limits, half-way moves	One noisy post cannot move the index; each move logs its headline
Interfaces	JSON-lines stream + CSV, FastAPI (/signals, /events, /analyze), Streamlit	Any downstream module, e.g. stress testing, can subscribe

KEY RESULTS

Measured against naive baselines

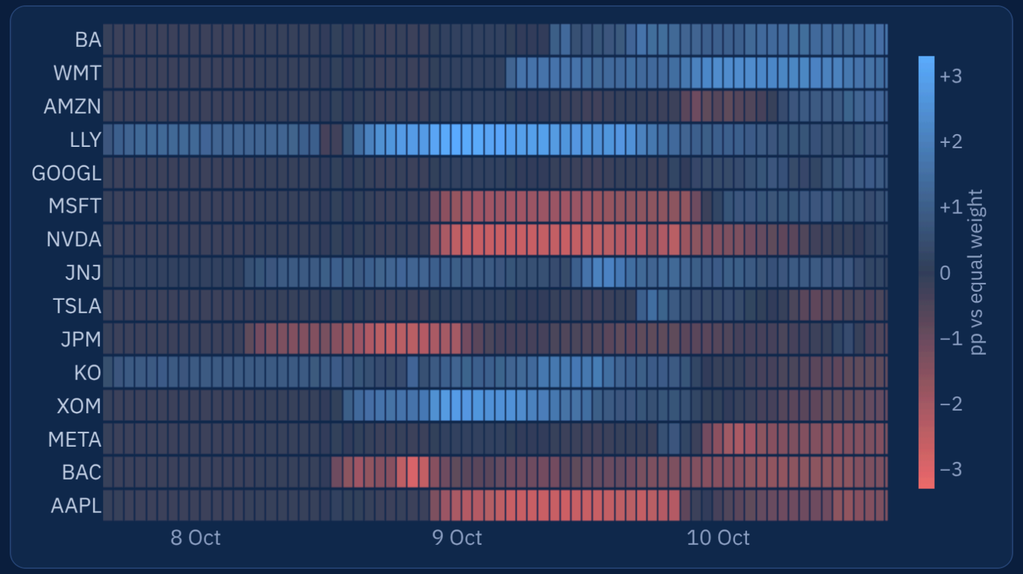
Task (test data)	Naive approach	RiskPulse
Sentiment, 2,388 labeled finance tweets	Generic lexicon: 49.4 %, F1 0.45	75.6 %, F1 0.71
Direction vs 117 StockTwits authors' tags	0.44 balanced accuracy	0.67 balanced accuracy
Event type, 60 held-out headlines	Always "General": 66.7 %	76.7 % accuracy
Reading ~650 headlines and posts	~3.6 analyst-hours	~30 s per cycle

569 signals

653 documents

74 rebalances

0.7 % avg. turnover



Index weights over 72 hours: blue = overweight, red = underweight. JPM was cut 0.53 pp after “Jamie Dimon warns the next credit crisis could be severe”.

DOMAIN IMPACT

Why it matters

Index & portfolio teams

A rules-based sentiment overlay that reacts within one cycle, with position limits, turnover control and an audit trail.

Risk & credit teams

Early warning on credit, geopolitical and macro events, scored 1–10 and served on /events, ready to trigger stress tests.

Governance

Every signal carries its keywords, model, source and link; every weight change names the headline behind it.

Cost & scale

Free data and CPU-only inference. A new stock is one line of config; a new source is one function.

From thousands of unread headlines to an explainable risk signal, every 15 minutes.

CHALLENGES, LIMITATIONS & NEXT STEPS

What we solved, what comes next

Challenges we solved

- GDELT allows 1 request per 5 s and rate-limits shared IPs → batched queries and snapshot fallback
- Noisy, promotional social posts → spam and copy-paste filter, 0.6× weight, impact cap of 7
- Feeds mix in loosely related stories → a company must be named in the headline
- Choosing a model on evidence → two transformers benchmarked on 2,388 labeled tweets

Limitations → next steps

- Sentiment is tone, not risk: “oil jumps on sanctions” reads positive → entity- and risk-aware sentiment
- Alias-based entity linking cannot tell subject from source → named-entity recognition
- Rule-based events (76.7 %) → fine-tune a classifier on the labeled headlines
- No price backtest yet → yfinance backtest vs equal weight; Module B on the /events feed